

38. The number of non-negative integral ordered pair(s) (x, y) for which $(xy - 7)^2 = x^2 + y^2$ holds is greater than or equal to :
 (a) 1 (b) 2 (c) 3 (d) 4
39. If $a < 0$, then the value of x satisfying $x^2 - 2a|x - a| - 3a^2 = 0$ is/are
 (a) $a(1 - \sqrt{2})$ (b) $a(1 + \sqrt{2})$ (c) $a(-1 - \sqrt{6})$ (d) $a(-1 + \sqrt{6})$
40. If $0 < a < b < c$ and the roots α, β of the equation $ax^2 + bx + c = 0$ are imaginary, then
 (a) $|\alpha| = |\beta|$ (b) $|\alpha| > 1$ (c) $|\beta| < 1$ (d) $|\alpha| = 1$
41. If x satisfies $|x - 1| + |x - 2| + |x - 3| > 6$, then
 (a) $x \in (-\infty, 1)$ (b) $x \in (-\infty, 0)$ (c) $x \in (4, \infty)$ (d) $(2, \infty)$
42. The value of k for which both roots of the equation $4x^2 - 2x + k = 0$ are completely in $(-1, 1)$, may be equal to :
 (a) -1 (b) 0 (c) 2 (d) -3
43. Let $\alpha, \beta, \gamma, \delta$ are roots of $x^4 - 12x^3 + \lambda x^2 - 54x + 14 = 0$
 If $\alpha + \beta = \gamma + \delta$, then
 (a) $\lambda = 45$ (b) $\lambda = -45$
 (c) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2$ then $\frac{\alpha\beta}{\gamma\delta} = \frac{7}{2}$ (d) If $\alpha^2 + \beta^2 < \gamma^2 + \delta^2 \Rightarrow \frac{\alpha\beta}{\gamma\delta} = \frac{2}{7}$
44. If $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right); \left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right); \left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ lie on $L: lx + my + n = 0$; where a, b, c are real numbers different from 1; then
 (a) $a + b + c = -\frac{m}{l}$ (b) $abc = \frac{m+n}{l}$
 (c) $ab + bc + ca = \frac{n}{l}$ (d) $abc - (ab + bc + ca) + 3(a + b + c) = 0$

Answers

1.	(a, c, d)	2.	(a, b, d)	3.	(a, c)	4.	(a, c, d)	5.	(a, b, c, d)	6.	(b, c)
7.	(a, b, c)	8.	(a, b, d)	9.	(a, c)	10.	(a, b, d)	11.	(a, b, c, d)	12.	(a, c)
13.	(b, d)	14.	(c, d)	15.	(a, b)	16.	(b)	17.	(b, c)	18.	(a, b, c)
19.	(a, b, d)	20.	(a, b, c, d)	21.	(a, c)	22.	(a, b, c, d)	23.	(a, b, d)	24.	(a, c, d)
25.	(a, c, d)	26.	(a, b)	27.	(b, c)	28.	(a, b)	29.	(a, b, d)	30.	(a, b, c, d)
31.	(d)	32.	(a, b)	33.	(a, c, d)	34.	(a, b, d)	35.	(a, b, d)	36.	(a, c)
37.	(b, c, d)	38.	(a, b, c, d)	39.	(a, d)	40.	(a, b)	41.	(b, c)	42.	(a, b)
43.	(a, c)	44.	(a, c, d)								

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $f(x) = ax^2 + bx + c$, $a \neq 0$, such that $f(-1-x) = f(-1+x) \forall x \in R$. Also given that $f(x) = 0$ has no real roots and $4a + b > 0$.

- Let $\alpha = 4a - 2b + c$, $\beta = 9a + 3b + c$, $\gamma = 9a - 3b + c$, then which of the following is correct ?
 (a) $\beta < \alpha < \gamma$ (b) $\gamma < \alpha < \beta$ (c) $\alpha < \gamma < \beta$ (d) $\alpha < \beta < \gamma$
- Let $p = b - 4a$, $q = 2a + b$, then pq is :
 (a) negative (b) positive (c) 0 (d) nothing can be said

Paragraph for Question Nos. 3 to 4

If α, β are the roots of equation $(k+1)x^2 - (20k+14)x + 91k+40 = 0$; $(\alpha < \beta) k > 0$, then answer the following questions.

- The smaller root (α) lie in the interval :
 (a) (4, 7) (b) (7, 10) (c) (10, 13) (d) None of these
- The larger root (β) lie in the interval :
 (a) (4, 7) (b) (7, 10) (c) (10, 13) (d) None of these

Paragraph for Question Nos. 5 to 7

Let $f(x) = x^2 + bx + c \forall x \in R$, ($b, c \in R$) attains its least value at $x = -1$ and the graph of $f(x)$ cuts y-axis at $y = 2$.

- The least value of $f(x) \forall x \in R$ is :
 (a) -1 (b) 0 (c) 1 (d) $3/2$
- The value of $f(-2) + f(0) + f(1) =$
 (a) 3 (b) 5 (c) 7 (d) 9
- If $f(x) = a$ has two distinct real roots, then complete set of values of a is :
 (a) $(1, \infty)$ (b) $(-2, -1)$ (c) $(0, 1)$ (d) $(1, 2)$

Paragraph for Question Nos. 8 to 9

Consider the equation $\log_2^2 x - 4\log_2 x - m^2 - 2m - 13 = 0$, $m \in R$. Let the real roots of the equation be x_1, x_2 such that $x_1 < x_2$.

- The set of all values of m for which the equation has real roots is :
 (a) $(-\infty, 0)$ (b) $(0, \infty)$ (c) $[1, \infty)$ (d) $(-\infty, \infty)$

9. The sum of maximum value of x_1 and minimum value of x_2 is :

- (a) $\frac{513}{8}$ (b) $\frac{513}{4}$ (c) $\frac{1025}{8}$ (d) $\frac{257}{4}$

Paragraph for Question Nos. 10 to 11

The equation $x^4 - 2x^3 - 3x^2 + 4x - 1 = 0$ has four distinct real roots x_1, x_2, x_3, x_4 such that $x_1 < x_2 < x_3 < x_4$ and product of two roots is unity, then :

10. $x_1x_2 + x_1x_3 + x_2x_4 + x_3x_4 =$
 (a) 0 (b) 1 (c) $\sqrt{5}$ (d) -1
11. $x_2^3 + x_4^3 =$
 (a) $\frac{2+5\sqrt{5}}{8}$ (b) -4 (c) $\frac{27\sqrt{5}+5}{4}$ (d) 18

Paragraph for Question Nos. 12 to 14

Let $f(x)$ be a polynomial of degree 5 with leading coefficient unity, such that $f(1) = 5, f(2) = 4, f(3) = 3, f(4) = 2$ and $f(5) = 1$, then :

12. $f(6)$ is equal to :
 (a) 120 (b) -120 (c) 0 (d) 6
13. Sum of the roots of $f(x)$ is equal to :
 (a) 15 (b) -15 (c) 21 (d) can't be determine
14. Product of the roots of $f(x)$ is equal to :
 (a) 120 (b) -120 (c) 114 (d) -114

Paragraph for Question Nos. 15 to 16

Consider the cubic equation in x , $x^3 - x^2 + (x - x^2)\sin\theta + (x - x^2)\cos\theta + (x - 1)\sin\theta\cos\theta = 0$ whose roots are α, β, γ .

15. The value of $\left(\frac{\alpha}{2}\right)^2 + \left(\frac{\beta}{2}\right)^2 + \left(\frac{\gamma}{2}\right)^2 =$
 (a) 1 (b) $\frac{1}{2}$
 (c) $2\cos\theta$ (d) $\frac{1}{2}(\sin\theta + \cos\theta + \sin\theta\cos\theta)$
16. Number of values of θ in $[0, 2\pi]$ for which at least two roots are equal, is :
 (a) 2 (b) 3 (c) 4 (d) 5

Paragraph for Question Nos. 17 to 18

Let $P(x)$ be a quadratic polynomial with real coefficients such that for all real x the relation $2(1 + P(x)) = P(x-1) + P(x+1)$ holds.

If $P(0) = 8$ and $P(2) = 32$ then :

17. The sum of all the coefficient of $P(x)$ is :
 (a) 20 (b) 19 (c) 17 (d) 15
18. If the range of $P(x)$ is $[m, \infty)$, then the value of m is :
 (a) -12 (b) 15 (c) -17 (d) -5

Paragraph for Question Nos. 19 to 21

Let t be a real number satisfying $2t^3 - 9t^2 + 30 - \lambda = 0$ where $t = x + \frac{1}{x}$ and $\lambda \in \mathbb{R}$.

19. If the above cubic has three real and distinct solutions for x then exhaustive set of value of λ be :
 (a) $3 < \lambda < 10$ (b) $3 < \lambda < 30$ (c) $\lambda = 10$ (d) None of these
20. If the cubic has exactly two real and distinct solutions for x then exhaustive set of values of λ be :
 (a) $\lambda \in (-\infty, 3) \cup (30, \infty)$ (b) $\lambda \in (-\infty, -22) \cup (10, \infty) \cup \{3\}$
 (c) $\lambda \in \{3, 30\}$ (d) None of these
21. If the cubic has four real and distinct solutions for x then exhaustive set of values of λ be :
 (a) $\lambda \in (3, 10)$ (b) $\lambda \in \{3, 10\}$
 (c) $\lambda \in (-\infty, -22) \cup (10, \infty)$ (d) None of these

Paragraph for Question Nos. 22 to 23

Consider a quadratic expression $f(x) = tx^2 - (2t-1)x + (5t-1)$

22. If $f(x)$ can take both positive and negative values then t must lie in the interval
 (a) $\left(\frac{-1}{4}, \frac{1}{4}\right)$ (b) $\left(-\infty, \frac{-1}{4}\right) \cup \left(\frac{1}{4}, \infty\right)$ (c) $\left(\frac{-1}{4}, \frac{1}{4}\right) - \{0\}$ (d) $(-4, 4)$
23. If $f(x)$ is non-negative $\forall x \geq 0$ then t lies in the interval
 (a) $\left[\frac{1}{5}, \frac{1}{4}\right]$ (b) $\left[\frac{1}{4}, \infty\right)$ (c) $\left[\frac{-1}{4}, \frac{1}{4}\right]$ (d) $\left[\frac{1}{5}, \infty\right)$

Answers

1. (c)	2. (a)	3. (a)	4. (c)	5. (c)	6. (d)	7. (a)	8. (d)	9. (d)	10. (b)
11. (d)	12. (a)	13. (a)	14. (c)	15. (b)	16. (d)	17. (b)	18. (c)	19. (c)	20. (b)
21. (a)	22. (c)	23. (d)							

Exercise-4 : Matching Type Problems

1.

(A)	The least positive integer x , for which $\frac{2x-1}{2x^3+3x^2+x}$ is positive, is equal to	(P)	$\frac{4}{3}$
(B)	If the quadratic equation $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possess roots of opposite sign then a can be equal to	(Q)	1
(C)	The roots of the equation $\sqrt{x+3} - 4\sqrt{x-1} + \sqrt{x+8} - 6\sqrt{x-1} = 1$ can be equal to	(R)	6
(D)	If the roots of the equation $x^4 - 8x^3 + bx^2 - cx + 16 = 0$ are all real and positive then $2(c - b)$ is equal to	(S)	16
		(T)	10

 2. Given the inequality $ax + k^2 > 0$. The complete set of values of 'a' so that

(A)	The inequality is valid for all values of x and k is	(P)	R
(B)	There exists a value of x such that the inequality is valid for any value of k is	(Q)	ϕ
(C)	There exists a value of k such that the inequality is valid for all values of x is	(R)	$\{0\}$
(D)	There exists values of x and k for which inequality is valid is	(S)	$R - \{0\}$
		(T)	$\{1\}$

3.

(A)	The real root(s) of the equation $x^4 - 8x^2 - 9 = 0$ are	(P)	No real roots
(B)	The real root(s) of the equation $x^{2/3} + x^{1/3} - 2 = 0$ are	(Q)	-3, 3
(C)	The real root(s) of the equation $\sqrt{3x+1} + 1 = \sqrt{x}$ are	(R)	-8, 1